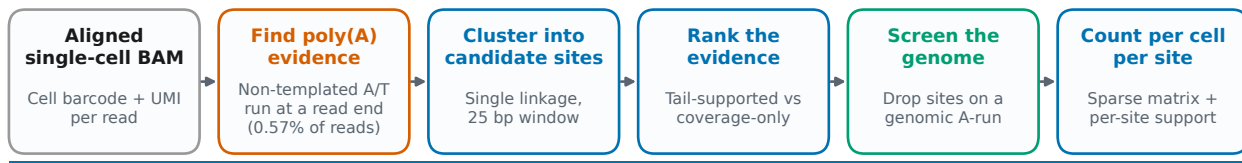


## a What PeakATail does, step by step



### peakatail run

One command produces the sites and the count matrix.

Two flags set the operating point: --polya-min-umis (tail molecules required) and --ip-filter (genome screen).



### Cell-type switch tests

`peakatail switch diff`

False-positive rate checked against shuffled labels

### 3'UTR length shifts

`peakatail switch length`

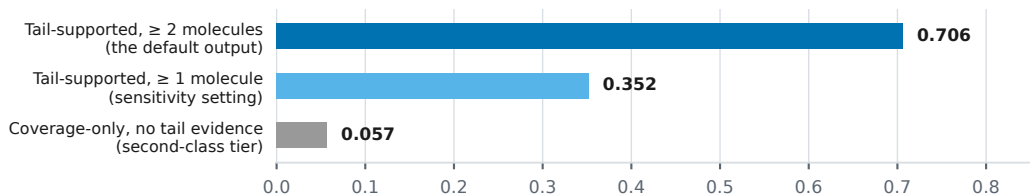
Per-cell PDUI along a differentiation axis

### Cross-sample replication

`replication_filter.py`

Keep a switch only if independent samples agree

## b Why a call is trusted: the evidence decides



### Atlas-agreement precision @100 bp

A site earns its place from molecules that carry the tail, not from a bump in coverage.

The three classes are reported separately, so a user chooses the operating point rather than inheriting ours.

PBMC 10k v3 shown. The default keeps 46,524 sites.

## c What the tool delivers

### Precision, four libraries

PBMC 10k v3 0.706  
Testis mouse 1 0.745  
Testis mouse 2 0.757  
PBMC donor 2 0.8279

Atlas-agreement precision @100 bp of the default output

### Statistics that hold

3.0% false-positive rate under shuffled labels  
0 of 20 null runs produced a hit

The switch test's error rate is measured, not assumed

### Runs on a laptop-class node

12.5 GB peak memory  
27 min for a 10k-cell library

Uncontended single run; no specialised high-memory node required

## d The result the tool is built for

**15,942 cell-type polyadenylation switches replicated across 12 patients of a lung-tumour cohort**

and none replicated in any of ten label-shuffle null runs of the same pipeline

A switch counts only when independent patients agree on it in the same direction, and any patient disagreeing vetoes the call.